

PROTOCOL DEVIATIONS ADDENDUM

Post-extraction classification and coding decisions

Protocol: PCNE protocol v4 (11 November 2025)

Decision dates: 18 and 28 August 2026

Status: Approved for the v10 codebook, data extraction, and analysis preparation

Scope: Post-extraction classification and coding rules; no change to study eligibility or author-reported data

1. Deviation and rationale

The registered protocol specifies offspring age in days and identifies age or age class as a possible biological moderator, but it does not prespecify species- and sex-specific thresholds for an age-class variable. This addendum records the threshold rule approved after data extraction and before effect-size modelling.

Primary analysis decision: offspring_age_d remains the continuous age moderator. The added offspring_age_class variable is restricted to descriptive summaries and sensitivity analyses. It will not replace offspring_age_d in the primary moderator model.

This is a classification decision rather than a correction to author-reported data. The reported or extracted offspring age remains unchanged.

2. Operational age classes

Thresholds follow the working postnatal stages summarized by Bell (2018). For the planned three-level variable, Bell's peripubertal and late-adolescent stages are combined as adolescent. Adult begins after the upper bound of Bell's late-adolescent stage.

Species	Sex	Juvenile	Adolescent	Adult
Rat	Female	PND 18-34	PND 35-70	PND >70
Rat	Male	PND 18-41	PND 42-70	PND >70
Mouse	Female	PND 18-27	PND 28-60	PND >60
Mouse	Male	PND 18-29	PND 30-60	PND >60

Source: Bell 2018, Table 1; DOI 10.1210/en.2018-00220.

Mixed-sex rule: classify a mixed-sex row only when the male and female rules give the same class at the recorded age. Otherwise code transitional_unclear and review it in sensitivity analysis.

Boundary rule: the present database contains no offspring ages below PND 18. If a future row falls below PND 18, do not force it into juvenile; record it for a new prespecified decision. Bell notes that exact timing can vary by strain and environmental context.

Application to PCN v10: 52 effect sizes are juvenile, 78 adolescent, and 77 adult. No current row is transitional_unclear.

3. Codebook implementation

The clean extraction file adds offspring_age_class immediately after offspring_age_d. It also renames gest_stage_ex_new to gest_window and uses the approved labels early_mid, mid_late, late, and full_gestation.

measurement_variable is retained unchanged because it identifies the specific behavioural outcome. No independent-arm indicator is added because the control and exposed animals are distinct for every comparison; repeated measurements and shared groups will instead be represented by clustering identifiers in the derived analysis data.

The original statistical names c_se and ex_se are restored for author-reported or directly extracted standard errors. Analysis-ready statistics use the _analysis suffix and contain the final values selected for modelling, including reproducible derivations where the author did not report the required statistic.

4. Standard-error provenance fields

c_se: Control-group SE reported by the authors or directly extracted from the source; preserved unchanged.

c_se_analysis: Final control-group SE selected for analysis. It equals c_se when reported and accepted; otherwise it is derived reproducibly.

c_se_derivation: Method used to obtain or validate c_se_analysis.

c_se_source: Text provenance for c_se_analysis: the exact paper table/figure, author file/email, or input source and derivation used. It is not a numeric SE and does not replace c_se.

The experimental-arm variables ex_se, ex_se_analysis, ex_se_derivation, and ex_se_source use the same definitions.

5. Control-condition uncertainty (Unclear)

Registered definition: The protocol defines control_conditions as the acoustic conditions under which control animals were kept during the maternal noise-exposure phase: silence, ambient sound, white noise, or other. It does not specify how to code a study when the control acoustic environment cannot be determined from the report.

Approved deviation: Add Unclear as an allowed value for control_conditions. Code Unclear when the article identifies a control or no-noise group but provides insufficient information to distinguish silence from ambient sound or another acoustic condition. This includes descriptions such as “no noise exposure,” “without audiogenic stimulation,” or equivalent handling in an exposure room or chamber when its background sound or acoustic isolation is not reported.

Do not use Unclear when: the source supports a more specific classification. Code `ambient_sound` when controls remained under stated standard or normal laboratory housing without experimental playback; code `silence` when controls were explicitly kept in an acoustically isolated, quiet, or noiseless environment without playback; code `white_noise` when white noise was explicitly used as the control sound; and code `other` for a reported acoustic condition that does not fit the preceding categories.

Documentation rule: For every Unclear value, `control_notes` must record the authors' description and the missing information that prevents a more specific classification. Leave `control_loudness` blank unless the control sound level was reported. Do not infer silence solely from the absence of experimental noise playback.

Rationale and impact: This category prevents uncertain reports from being forced into silence or ambient sound and makes the uncertainty available for descriptive summaries and moderator analyses. It does not broaden the comparator eligibility criterion: eligible controls must still represent dams not exposed to the experimental chronic-noise treatment. The change harmonizes the registered definition with the operational moderator dictionary and the reviewed included-study evidence.

6. References retained for citation

Bell, M. R. (2018). Comparing postnatal development of gonadal hormones and associated social behaviors in rats, mice, and humans. *Endocrinology*, 159(7), 2596-2613. <https://doi.org/10.1210/en.2018-00220>

Moore, E. M., Linsenbardt, D. N., Melón, L. C., & Boehm, S. L., II. (2011). Ontogenetic differences in adolescent and adult C57BL/6J and DBA/2J mice: Anxiety-like, locomotor, and consummatory behaviors. *Developmental Psychobiology*, 53(2), 141-156. <https://doi.org/10.1002/dev.20501>

Bell (2018) is the threshold authority for the operational rule. Moore et al. (2011) is retained as supporting context for mouse adolescence and behavioural testing across development.